Day-04

Short Questions

Question 1: Matrix Dimension in SVD.

You are given a dataset X representing 1000 user and 50 features. You compute the full singular value decomposition: $X = U\Sigma V^T$.

What are the exact geometric dimension (rows x cols) of the matrices U, Σ and V^T?

Solution: Full SVD shapes

U is 1000×1000,

Σ is 1000×50,

V^T is 50×50.

Question 2: Truncated SVD and Data compression.

You deide to perform a low-rank approximation of your 1000×50 dataset by keeping only the

top K=5 singular values.

A) what is the shape of the truncated Σ_K matrix?

B) why do we specially keep the largest singular value and discard the smallest ones? what physical property of the data do the largest singular values represent?

<u>Solution:</u>

A) the truncated Σ_K matrix is: 5X5

B) we keep the largest singular values because they correspond to the directions of max variance in the dataset.

They represent the strongest pattern, while the smallest singular values represent tiny, random Noise.

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Question 3: SVD vs eigendecomposition

You have a square, symmetric covariance matrix C. You can decompose it using either eigendecomposition ($C = Q \Lambda Q^T$) or ($C = V \Sigma V^T$)

For specially symmetric, positive semidefinite matrix like a covariance matrix, What is the exact relationship between the matrics Q an U, Between Λ and Σ?

Solution: For symmetric, positive semidefinite matrix, SVD and eigen decomposition are mathmaticly same. Therefore $Q = U = V$ and the eigenvalues perfectly equal the singular values ($\Lambda = \Sigma$).

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Question 4: PCA under the hood.

you call from sklearn.decomposition import PCA and run: PCA.fit(X). Under the hood, sklearn doesn't compute the covariance matrix $X^T X$, because doing so squares the condition number and causes floating point precision loss. Instead, it computes the SVD of the centered matrix X.

Which matrix from the SVD output (U, Σ or V^T) directly contains the principle components, that sklearn returns you?

Solution:

The matrix V^T (the right singular vector) directly contains the principle components. It now represent the axes of maximum variance.

Question 5: The Condition Number & Collinearity

The Condition Number of a matrix is the ratio of its largest singular value to its smallest non-zero singular value ($\sigma_{max}/\sigma_{min}$).

If you have a dataset where two features are perfectly correlated, what will the smallest singular value (σ_{min}) be? What happens to the condition number, and what does this tell you about the stability of a Linear Regression model trained on this data?

Solution: If two features are perfectly correlated, the matrix loses full rank, and the smallest singular value will be exactly $\sigma_{min}=0$. The condition number ($\sigma_{max}/0$) approaches infinity.

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This tell you the system is perfectly singular, and standard linear regression will fail unless regularization technique like Ridge/lasso are applied.

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